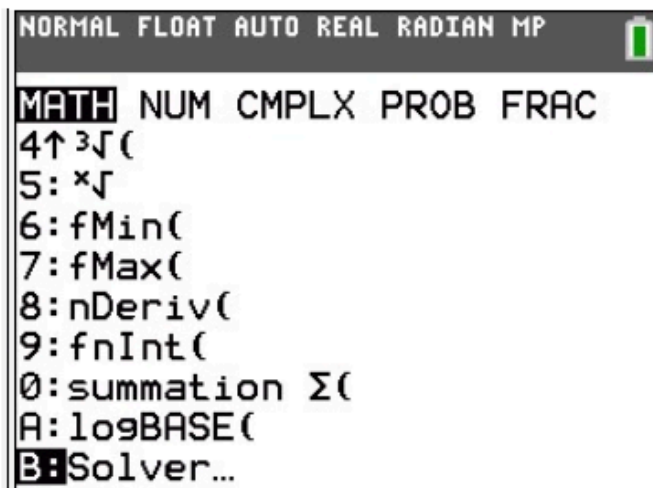
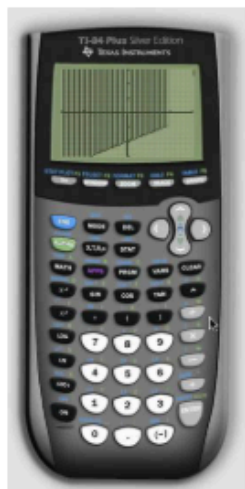


Calculus II Lesson 36: Arc Length of Polar Curve

TI-84 Calculator fnInt Feature



Numerical Integration on TI-84:

Press [MATH]

Select fnInt

Input the function, variable, and lower/upper bounds

Example: To find $\int_0^2 x^2 \cdot dx$

Input fnInt(X^2, X, 0, 2). for older TI-84

Input $\int_0^2 (x^2) dx$ for newer TI-84

1) Find a polar equation for the parabola with its focus at the pole.

Parabola has: $e = \text{eccentricity} = 1$; Directrix: $x = -2$

a) focus = _____

b) vertex = _____

c) Find distance from focus at the pole to directrix $= |d| =$ _____ ?

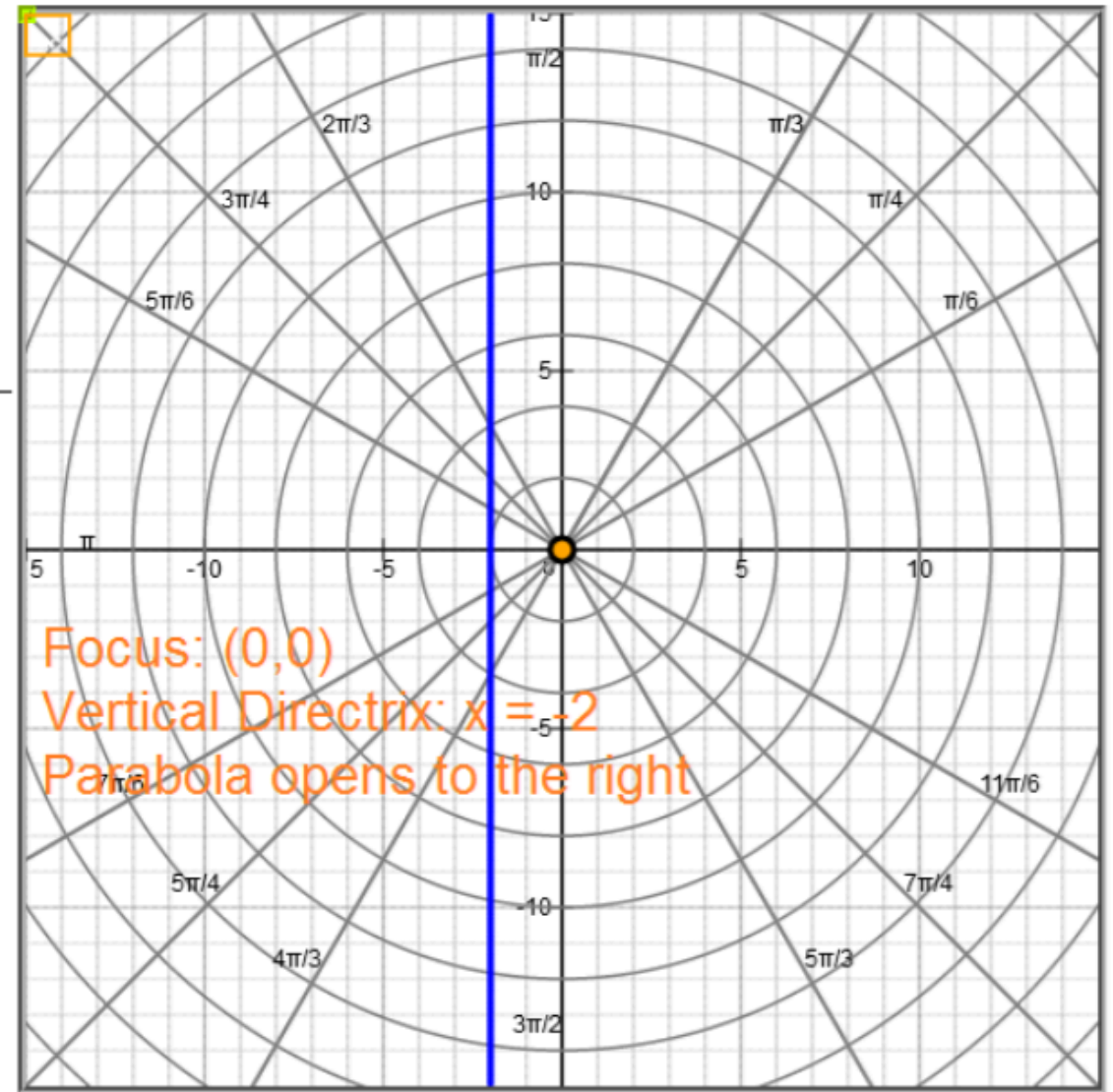
d) Polar equation for parabola: _____ ? (multiple choice)

A) $r = \frac{2}{1 + \cos \theta}$

B) $r = \frac{2}{1 - \cos \theta}$

C) $r = \frac{-2}{1 + \cos \theta}$

D) $r = \frac{-2}{1 - \cos \theta}$



2) Find a polar equation for the parabola with its focus at the pole.
Parabola has vertex at $(1, \pi/2)$.

a) focus = _____ b) Equation of directrix: _____

c) Find distance from focus at the pole to directrix $= |d| = \underline{\hspace{1cm}} ?$

d) e = eccentricity = _____ Hint: Conic is a parabola.

e) Polar equation for parabola: _____ ? (multiple choice)

A) $r = \frac{1}{1 + \sin \theta}$

B) $r = \frac{2}{1 + \sin \theta}$

C) $r = \frac{3}{1 - \sin \theta}$

D) $r = \frac{4}{1 - \sin \theta}$

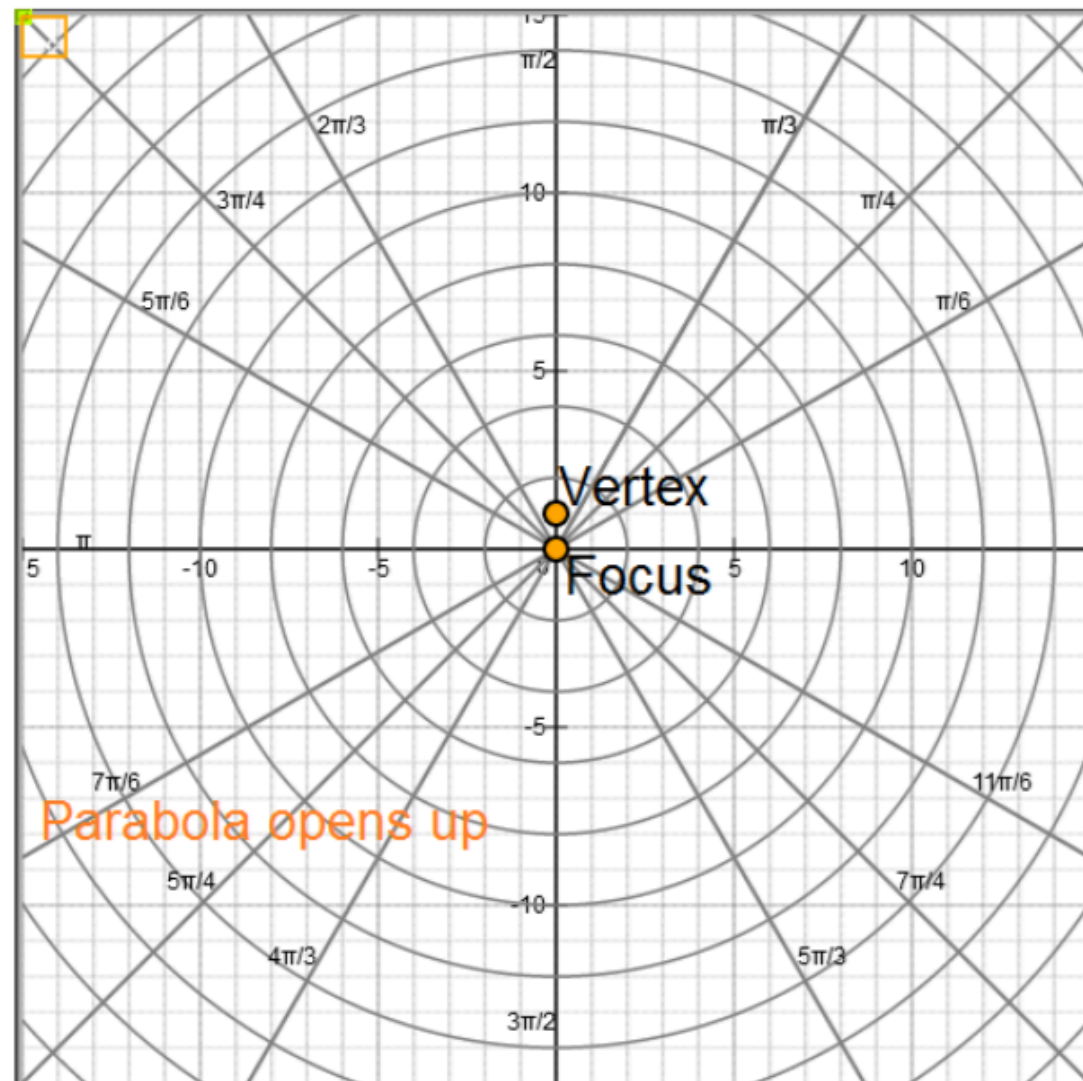
Four types of equations based on location and type of directrix:

1) Horizontal directrix above the pole: $r = \frac{ed}{1 + e \sin \theta}$

2) Horizontal directrix below the pole: $r = \frac{ed}{1 - e \sin \theta}$

3) Vertical directrix to the right of the pole: $r = \frac{ed}{1 + e \cos \theta}$

4) Vertical directrix to the left of the pole: $r = \frac{ed}{1 - e \cos \theta}$



3) Find a polar equation for the ellipse with its focus at the pole.

Ellipse has: $e = \text{eccentricity} = 1/3$; Directrix: $y = 3$

a) focus = _____

b) Find distance from focus at the pole to directrix $= |d| =$ _____ ?

c) Polar equation for ellipse: _____ ?

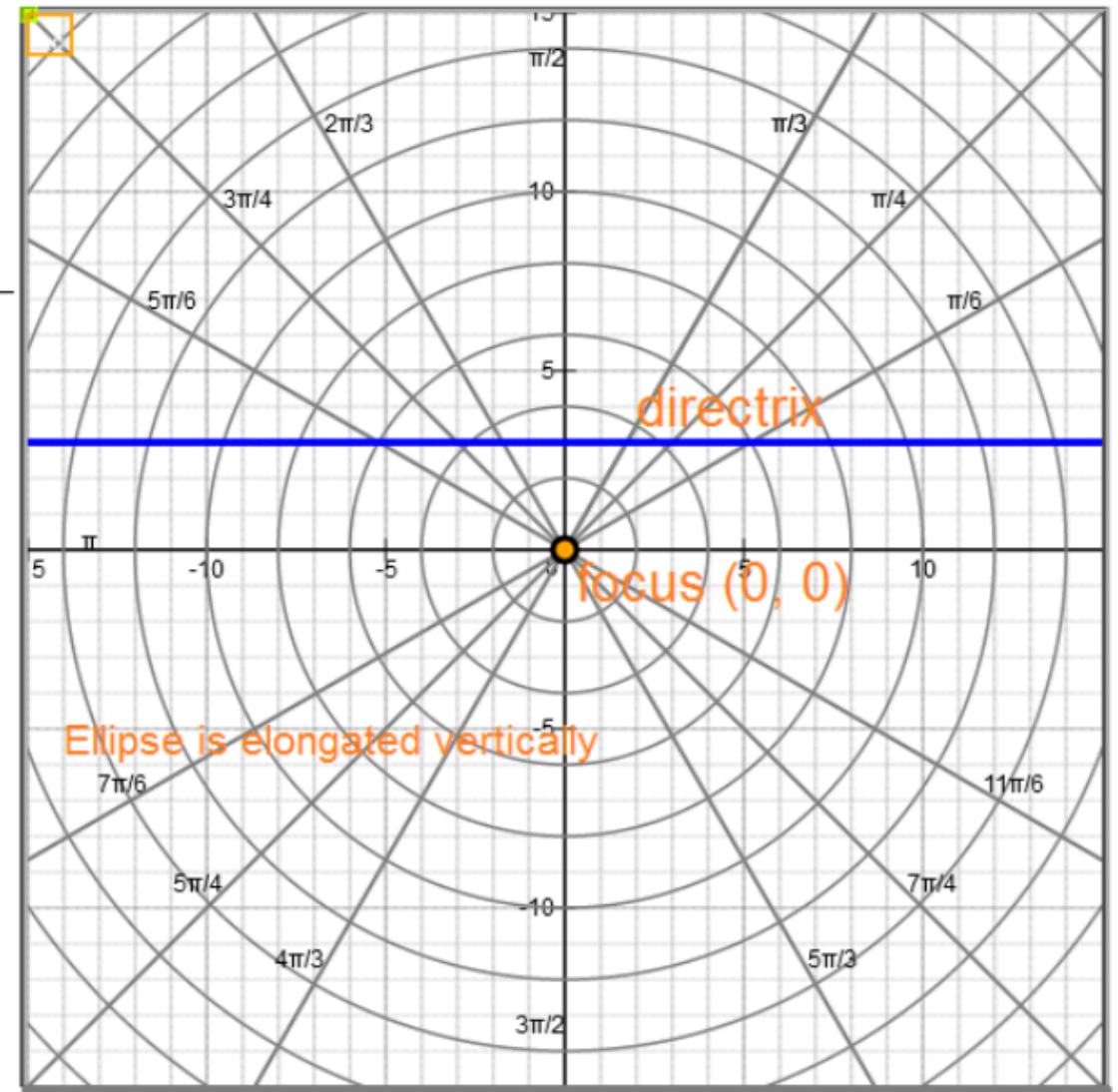
Four types of equations based on location and type of directrix:

1) Horizontal directrix above the pole: $r = \frac{ed}{1 + e \sin \theta}$

2) Horizontal directrix below the pole: $r = \frac{ed}{1 - e \sin \theta}$

3) Vertical directrix to the right of the pole: $r = \frac{ed}{1 + e \cos \theta}$

4) Vertical directrix to the left of the pole: $r = \frac{ed}{1 - e \cos \theta}$



4) Find a polar equation for the ellipse with its focus at the pole.

Ellipse has vertices at $(2, 0)$ and $(6, \pi)$.

a) focus = ? b) center of ellipse = ?

c) a = distance from center to vertex = ?

d) c = distance from center to focus = ?

e) e = eccentricity = c/a = ?

f) Find distance from focus at the pole to directrix = $|d|$ = ?

Hint: Equation of ellipse will have the form: $r = \frac{ed}{1 \pm e \cos \theta}$

Use the point $(r = 2, \theta = 0)$ to find d .

g) Polar equation for ellipse: ?

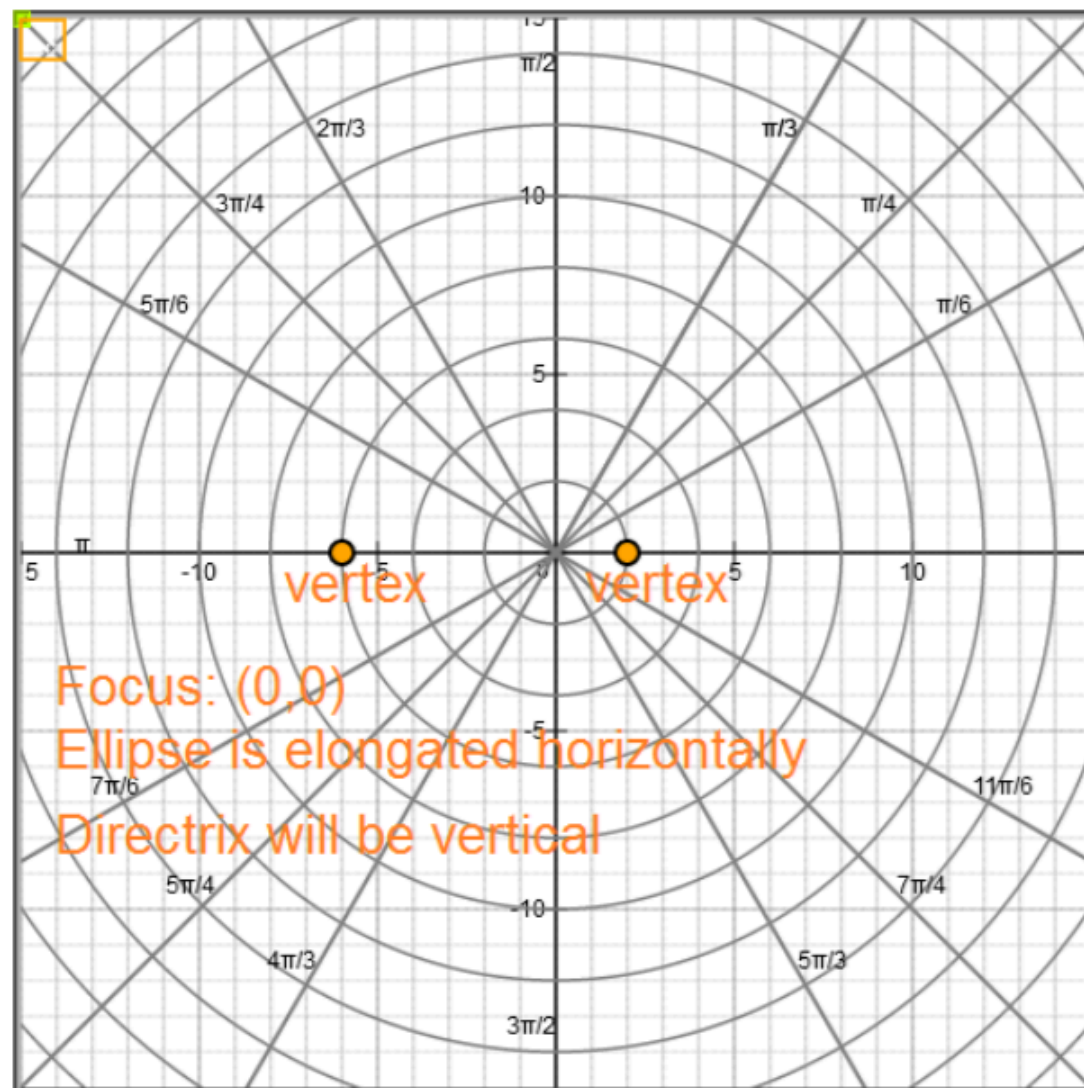
Four types of equations based on location and type of directrix:

1) Horizontal directrix above the pole: $r = \frac{ed}{1 + e \sin \theta}$

2) Horizontal directrix below the pole: $r = \frac{ed}{1 - e \sin \theta}$

3) Vertical directrix to the right of the pole: $r = \frac{ed}{1 + e \cos \theta}$

4) Vertical directrix to the left of the pole: $r = \frac{ed}{1 - e \cos \theta}$



5) Find a polar equation for the hyperbola with its focus at the pole.

Hyperbola has: $e = \text{eccentricity} = 4$; Directrix: $x = 4$

a) Find distance from focus at the pole to directrix = $d =$ _____ ?

b) Polar equation for hyperbola: _____ ?

Note: Directrix should be between the two branches of the hyperbola.

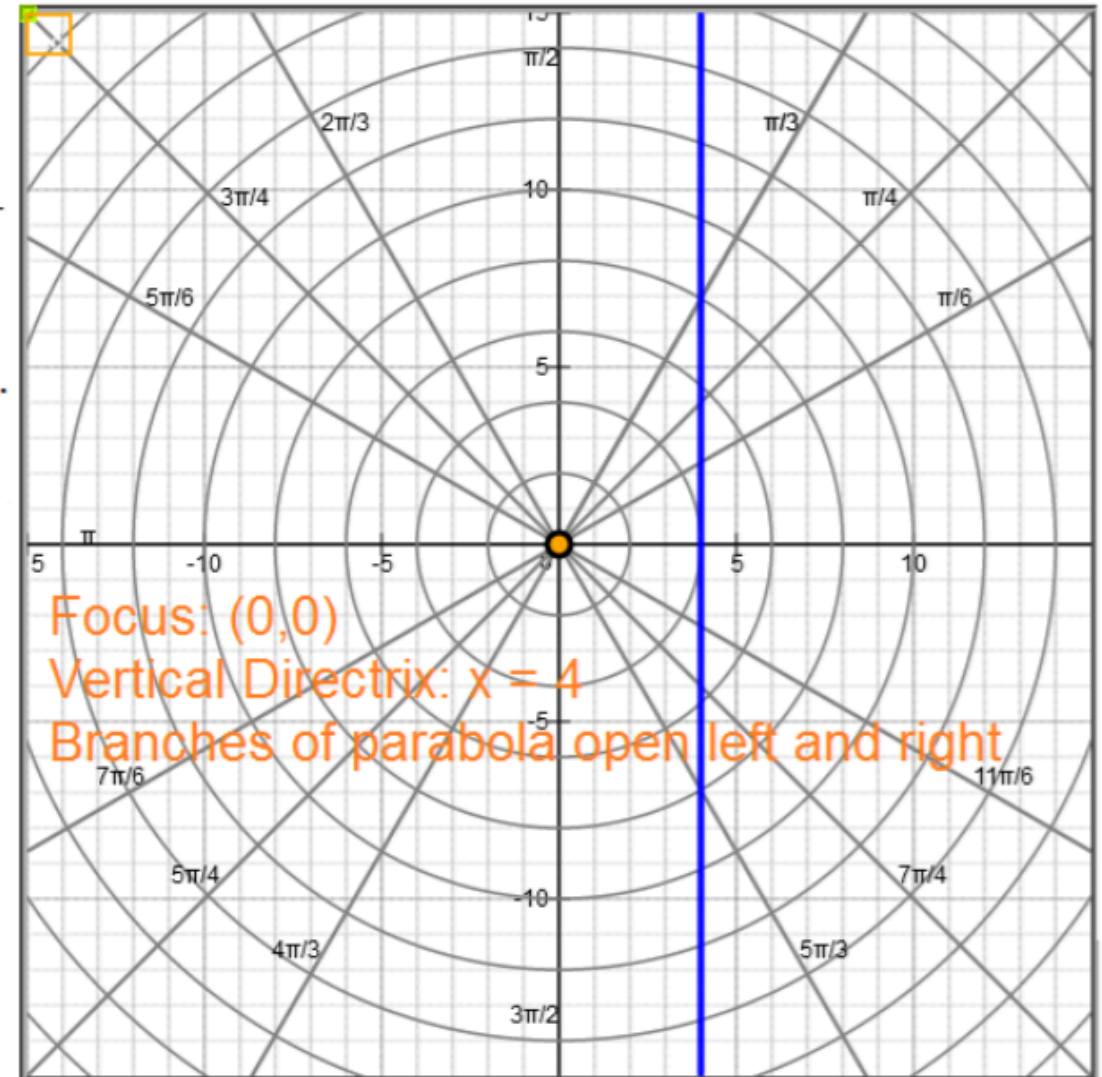
Four types of equations based on location and type of directrix:

1) Horizontal directrix above the pole: $r = \frac{ed}{1 + e \sin \theta}$

2) Horizontal directrix below the pole: $r = \frac{ed}{1 - e \sin \theta}$

3) Vertical directrix to the right of the pole: $r = \frac{ed}{1 + e \cos \theta}$

4) Vertical directrix to the left of the pole: $r = \frac{ed}{1 - e \cos \theta}$



6) Find a polar equation for the hyperbola with its focus at the pole.

Hyperbola has vertices at $(1, 3\pi/2)$ and $(7, 3\pi/2)$.

a) focus = ?

b) center of hyperbola = ?

c) a = distance from center to vertex = ?

d) c = distance from center to focus = ?

e) e = eccentricity = c/a = ?

f) Find distance from focus at the pole to directrix = $|d|$ = ?

Hint: Equation of hyperbola will have the form: $r = \frac{ed}{1 \pm e \sin \theta}$.

Use the point $(r = 1, \theta = 3\pi/2)$ to find d .

g) Polar equation for hyperbola: ?

Note: Directrix should be between the two branches of the hyperbola.

Four types of equations based on location and type of directrix:

1) Horizontal directrix above the pole: $r = \frac{ed}{1 + e \sin \theta}$

2) Horizontal directrix below the pole: $r = \frac{ed}{1 - e \sin \theta}$

3) Vertical directrix to the right of the pole: $r = \frac{ed}{1 + e \cos \theta}$

4) Vertical directrix to the left of the pole: $r = \frac{ed}{1 - e \cos \theta}$

